

■ B.7.4 Pipes

You can use pipes in place of files in *Mathematica* input-output functions. *Mathematica* uses the rule that any “file name” beginning with a `!` is in fact a pipe. The remainder of the name is used to specify the external command to which the pipe is attached. When the external command contains spaces or other special characters, you have to enclose the whole name in double quotes.

The text you give after `!` is passed to your operating system or shell for execution. Under UNIX, external programs will be sought using whatever UNIX `PATH` you have specified. You should realize that this `PATH` has no connection with the `$Path` variable that you use within *Mathematica*. Nevertheless, the `math` shell script often resets your UNIX `PATH` so as to include directories of *Mathematica* utility programs.